

Exercise: Work with the Group Object

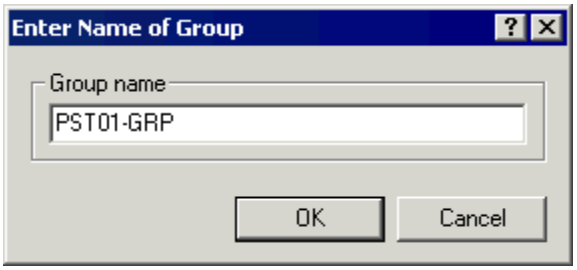
Create a Family of Linked Group Objects

Purpose

Learn how to create a family of linked group objects and how to update the group objects simultaneously and dynamically.

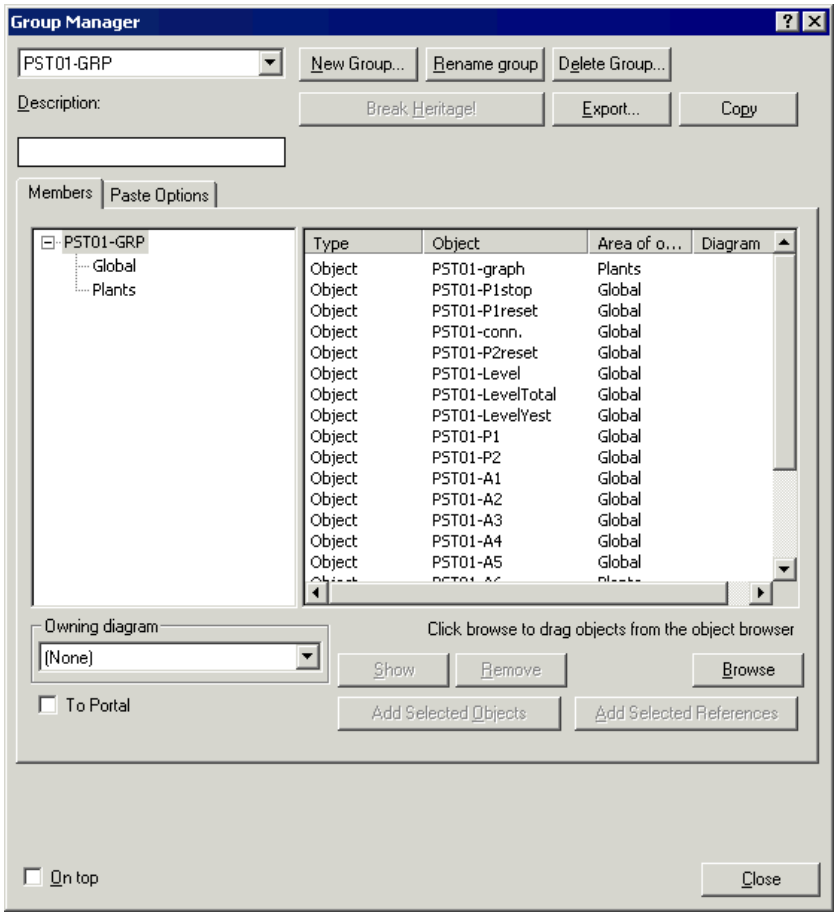
For this exercise, we use the IGSS Demo project.

Task 1: Create three linked Groups and show them on the same diagram

Step	Action
1	In **Definition** go to the **File** menu and near the bottom click on the **demo.elm** file to open the Demo project. In the **Plants** area, open the **Wastewater treatment** diagram and click on the **Pump Station** button.
2	Select all objects on the diagram.
3	In the **Edit** menu, select **Group Manager** and create a new group called `PST01-GRP`. 

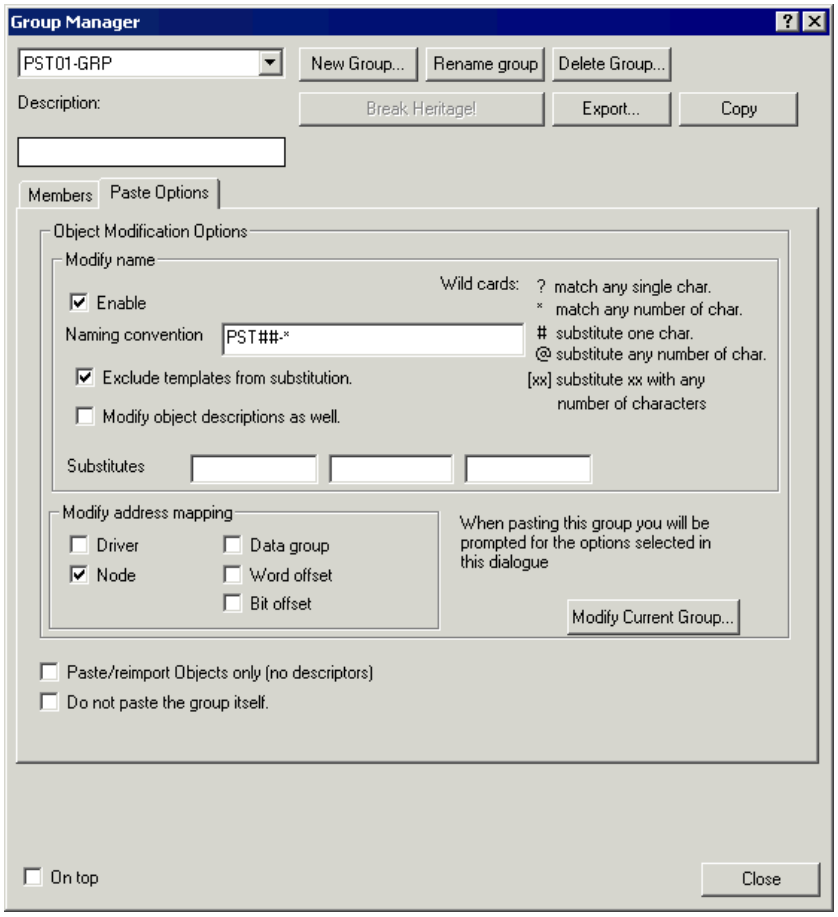


4	On the **Members** tab, click the **Add Selected Objects** button.

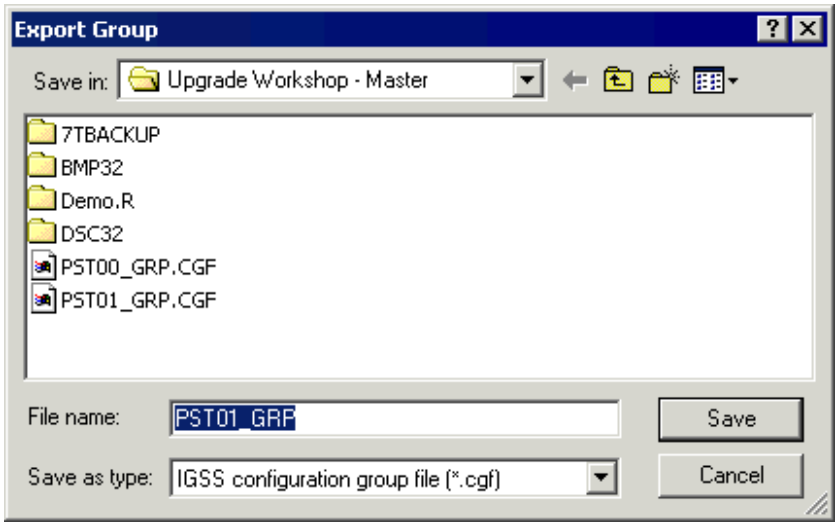
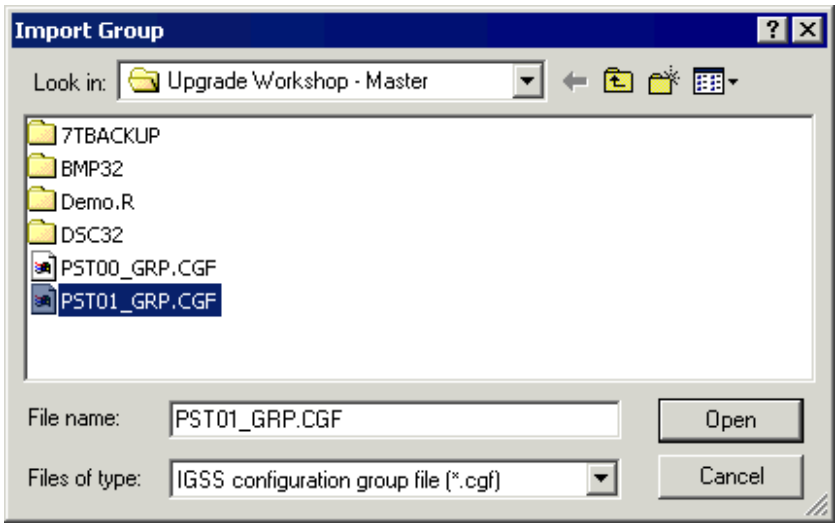




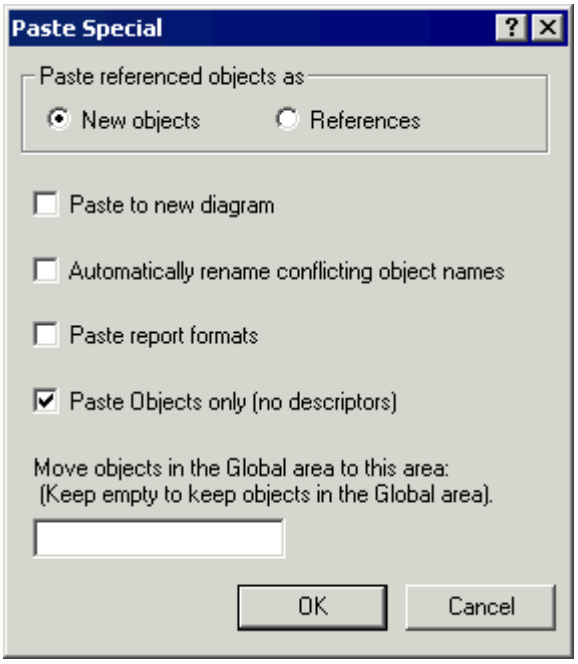
5	On the **Paste Options** tab, do the following: • Under **Modify name**, select the **Enable** check box. • In the **Naming convention** field, type `PST##-*` • Select the **Exclude templates from substitution** check box • Under **Modify address mapping,** select the **Node** check box



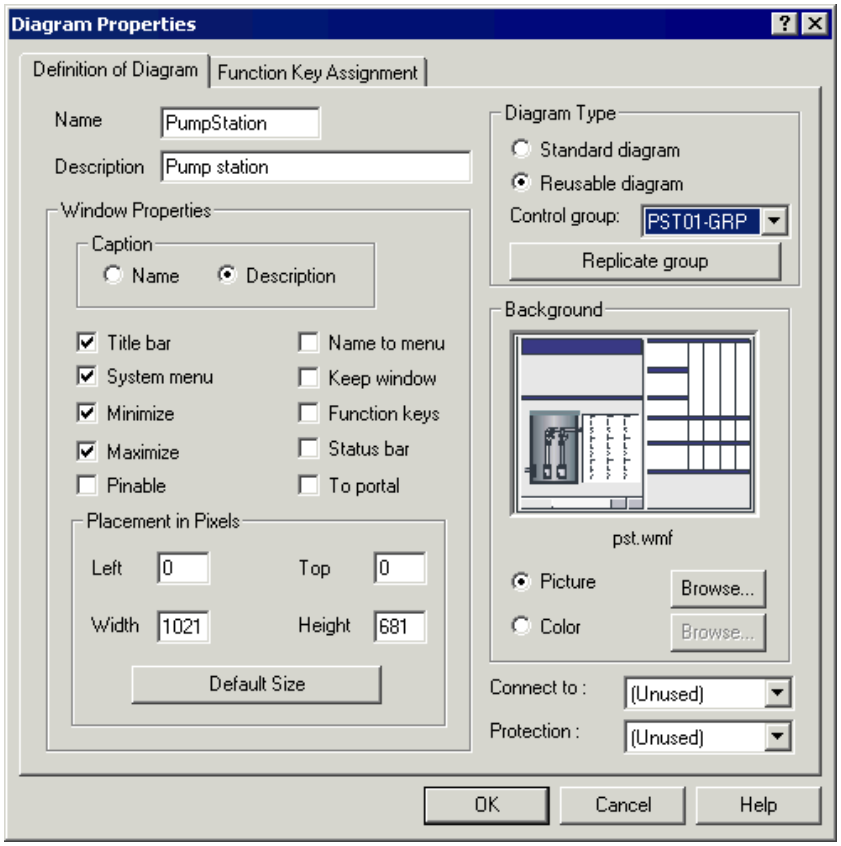


6	Click the **Export** button and name the file `PST01-GRP.cgf`. 
7	In the **File** menu, select **Import**. 

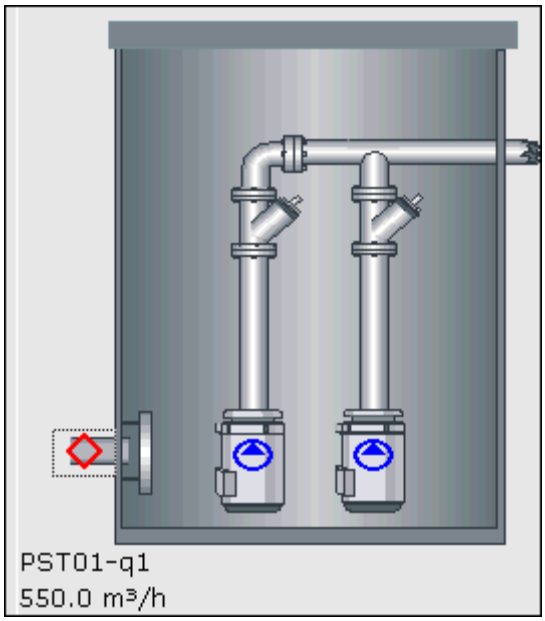


8	In the **Paste Special** dialogue, do the following: • Make sure that **New objects** is selected • Select the **Paste objects only** check box • Clear all other check boxes • Click **OK**. 
9	In the **Modify Options of Group Members** dialogue, do the following: • In the **Substitutes** field, type `02`. • In the **Node** field, type `#2` to assign node no. 2 to all objects in the group.
10	Repeat steps 7 – 9 now using `03` as object name substitute and `#3` as node substitute.
11	Right-click the diagram and select **Properties**.



12	Do the following: • Select the **Reusable diagram** option • Select **PST01-GRP** as control group • Click **OK**. 
13	Press F5 to refresh the diagram. *RESULT:* The three groups now appear. Select **PST02-GRP** and see how the object names change on the diagram.
14	Press F5 again and this time select **PST01-GRP**.



15	Add a new object to the diagram named `PST01-q1` to match the name convention. Place it as shown below. 
16	Select the symbol and activate the **Group Manager** dialogue.
17	Select the group **PST01-GRP** and click the **Add Selected Objects** button.
18	Click the **Export** button and overwrite the existing file, **PST01-GRP.cgf**.
19	In the **File** menu, select **Reimport**. All three groups will be selected. Accept the default settings and click **OK**.
20	Press F5 to refresh the diagram and select **PST02-GRP**. *RESULT:* The new object **PST02-q1** appears at the position where you placed the new object. The same goes for the group **PST03-GRP** where **PST03-q1** has been created.

YOU HAVE SUCCESSFULLY COMPLETED THE EXERCISE.